

human

Show me all the bridges and types of bridges in Exeter.

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to see all bridges and their types in Exeter. Provide the general steps to solve this geospatial query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are bridges, and we need to find all types of bridges in Exeter.

output steps: ["Find Exeter 1 area", "Search for all bridges in Exeter as many search results", "Identify and categorize the types of bridges found in Exeter"]

host_agent

Tool send_message returned result : The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are bridges, and we need to find all types of bridges in Exeter.

output steps: ["Find Exeter 1 area", "Search for all bridges in Exeter as many search results", "Identify and categorize the types of bridges found in Exeter"]

Calling tool generate_metadata_for_all_artifacts with args : {} Tool generate_metadata_for_all_artifacts returned result : {} Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,\n count of records fetched : 98.\n ',]

host_agent

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['description'].str.lower().isin(['city', 'town', 'suburban area', 'part of settlement', 'village'])) |
                    (gdf['descriptiongroup'].str.lower() == 'settlement')) &
                  (gdf['name1_text'].str.lower().str.contains('exeter'))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon filtered to represent the area of Exeter, using name and settlement-related columns.'
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 1 polygon for Exeter. The artifact 'filtered_exeter_area_polygon' contains the polygon(s) from 'exeter_area_polygon' that represent the area of Exeter, using both the name and settlement-related columns for accurate filtering.

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'.

host_agent

Calling tool send_message with args : {'target': 'structures_agent', 'task_description': "Search for all bridges and their types within the artifact 'filtered_exeter_area_polygon'."}

structures_agent

Calling tool call_os_ngd with args : {'filters': ['Bridge', 'Footbridge', 'Lift Bridge', 'Swing Bridge', 'Transporter Bridge'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_bridges'} Tool call_os_ngd returned result : ['Multiple search results have been found in

multiple datasets. A summary of each is provided.\n Artifacts generated are : ['exeter_bridges'],\n Descriptions are : ["A geopandas dataframe containing compound structure data with filters and bbox applied for the query Search for all bridges and their types within the artifact 'filtered_exeter_area_polygon'. using filters ['Bridge', 'Footbridge', 'Lift Bridge', 'Swing Bridge', 'Transporter Bridge']. Further Name filtering is available for this"],\n counts of records fetched are : [182].\n ', []

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Please analyze the dataset 'exeter_bridges' to identify and list all types of bridges found within the 'filtered_exeter_area_polygon'. Save the filtered results as 'filtered_exeter_bridges'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_bridges', 'filtered_exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def filter_and_list_bridge_types(data: list):
    import geopandas as gpd
    # exeter_bridges = data[0], filtered_exeter_area_polygon = data[1]
    bridges = data[0]
    area = data[1]
    # Spatial filter: bridges within the area polygon
    filtered = bridges[bridges.geometry.within(area.geometry.iloc[0])].copy()
    # List all unique bridge types in the filtered set
    bridge_types = filtered['description'].dropna().unique().tolist()
    # Save filtered results as 'filtered_exeter_bridges'
    return [
        f"Filtered {len(filtered)} bridges within the Exeter area. Bridge types found: {bridge_types}",
        'filtered_exeter_bridges',
        'GeoDataFrame of bridges within the filtered Exeter area polygon, including all attributes and geometry. Also includes the types of bridges found in the',
        filtered
    ]
```

Filtered 175 bridges within the Exeter area. Bridge types found: Bridge, Footbridge, and Swing Bridge.

A GeoDataFrame artifact named 'filtered_exeter_bridges' has been created, containing all attributes and geometry for bridges within the filtered Exeter area polygon, along with the types of bridges found in the area listed in the 'description' column.

structures_agent

I found 175 bridges within the 'filtered_exeter_area_polygon'. The types of bridges identified are:

- Bridge
- Footbridge
- Swing Bridge

The filtered results have been saved as 'filtered_exeter_bridges'.

host_agent

Calling tool send_message with args : {'target': 'land_features', 'task_description': "Search for all bridges and their types within the artifact 'filtered_exeter_area_polygon'."}

land_features_agent

I cannot solve this query as it involves searching for features associated with structures, specifically bridges, which are not part of the land features database.

host_agent

Calling tool send_message with args : {'target': 'water_network', 'task_description': "Search for all bridges and their types within the artifact 'filtered_exeter_area_polygon'."}

water_network_agent

I cannot assist with searching for bridges and their types. My capabilities are focused on water network features like rivers, streams, lakes, lochs, drains, and canals. If you have any queries related to these, feel free to ask!

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot all bridges and their types from the artifact 'filtered_exeter_bridges' within the area of 'filtered_exeter_area_polygon'. Include a legend or labels to distinguish bridge types (Bridge, Footbridge, Swing Bridge)."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['filtered_exeter_bridges', 'filtered_exeter_area_polygon']} Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def plot_bridges_by_type_in_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping
    import branca

    # Unpack data
    bridges = data[0].copy()
    area = data[1].copy()

    # Ensure CRS is the same and convert to EPSG:4326 for folium
    if bridges.crs != area.crs:
        bridges = bridges.set_crs(area.crs, allow_override=True)
    if bridges.crs != 'EPSG:4326':
        bridges = bridges.to_crs('EPSG:4326')
        area = area.to_crs('EPSG:4326')

    # Filter bridges within the area polygon
    area_poly = area.iloc[0].geometry
    bridges_in_area = bridges[bridges.intersects(area_poly)]

    # Only keep relevant bridge types
    bridge_types = ['Bridge', 'Footbridge', 'Swing Bridge']
    bridges_in_area = bridges_in_area[bridges_in_area['description'].isin(bridge_types)]

    # Color mapping for bridge types
    color_dict = {'Bridge': 'blue', 'Footbridge': 'green', 'Swing Bridge': 'red'}

    # Create folium map centered on area
    centroid = area_poly.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

    # Add area polygon
    folium.GeoJson(
        mapping(area_poly),
        name='Exeter Area',
        style_function=lambda x: {'fillColor': 'gray', 'color': 'black', 'weight': 2, 'fillOpacity': 0.1}
    ).add_to(m)
```

```

# Add bridges by type
for btype in bridge_types:
    sub = bridges_in_area[bridges_in_area['description'] == btype]
    for idx, row in sub.iterrows():
        name = row['name1_text'] if row['name1_text'] else btype
        folium.GeoJson(
            mapping(row['geometry']),
            name=f"{btype}",
            style_function=lambda x, color=color_dict[btype]: {'color': color, 'weight': 4, 'fillOpacity': 0.5},
            tooltip=folium.Tooltip(f"{name} ({btype})")
        ).add_to(m)

# Add legend
legend_html = '''
<div style="position: fixed; \n        bottom: 50px; left: 50px; width: 180px; height: 110px; \n        border:2px solid grey; z-index:9999; font-size:14px; bac
&nbsp;<b>Bridge Types</b><br>
&nbsp;<span style='color:blue;'>#9632;</span> Bridge<br>
&nbsp;<span style='color:green;'>#9632;</span> Footbridge<br>
&nbsp;<span style='color:red;'>#9632;</span> Swing Bridge<br>
</div>
'''
m.get_root().html.add_child(branca.element.Element(legend_html))

# Save map
map_filename = 'exeter_bridges_by_type_map.html'
m.save(map_filename)

# Prepare summary
summary = f"Plotted {len(bridges_in_area)} bridges in Exeter area. First 5:\n" + str(bridges_in_area[['name1_text', 'description']].head(5))
artifact_name = 'exeter_bridges_by_type_map'
artifact_description = 'Folium map showing all bridges (Bridge, Footbridge, Swing Bridge) in Exeter area, colored by type with legend.'
return [summary, artifact_name, artifact_description, map_filename]

```

['Plotted 175 bridges in Exeter area. First 5:\n name1_text description\n0 None Bridge\n1 None Bridge\n3 None Bridge\n4 None Bridge\n5 None Bridge', 'exeter_bridges_by_type_map', 'Folium map showing all bridges (Bridge, Footbridge, Swing Bridge) in Exeter area, colored by type with legend.', 'exeter_bridges_by_type_map.html']

host_agent

Here is the result for your query about bridges in Exeter:

- All bridges within the Exeter area have been identified and mapped.
- The types of bridges found are:

- Bridge
- Footbridge
- Swing Bridge

A map has been generated showing all 175 bridges in Exeter, with each type of bridge color-coded and a legend included for easy identification.

If you would like to view or download the map, let me know! If you need more details about specific bridges or want to filter by type, I can assist further.